CHAPTER

2

DNA Extraction Methods



PURPOSE OF DNA EXTRACTION

A biological sample obtained from a crime scene in the form of a blood or semen stain or a tissue (blood or buccal swab) sample from a known individual contains a number of substances besides DNA. Cellular proteins that package and protect DNA in the environment of the cell can inhibit the ability to analyze the DNA. Therefore, extraction methods have been developed to separate proteins and other cellular materials from the DNA molecules. Ideally, the DNA extraction process removes inhibitors that reduce or prevent polymerase chain reaction (PCR) amplification. The extraction process should also produce a stable solution containing high-quality DNA that will not degrade over time during sample storage.

The goals of the DNA extraction process are typically to (1) lyse cells to release the DNA molecules, (2) separate the DNA molecules from other cellular material, and (3) isolate the DNA into a format compatible with downstream applications including PCR amplification. The quantity and quality of DNA often need to be measured prior to proceeding further with analytical procedures to ensure optimal results. This chapter will focus on the DNA extraction process while the following chapter will cover assessment of DNA quantity and quality prior to further sample processing.

All samples must be carefully handled regardless of the DNA extraction method to avoid sample-to-sample contamination or introduction of extraneous DNA. The extraction process is probably where the DNA sample is more susceptible to contamination in the laboratory than at any other time in the forensic DNA analysis process. For this reason, laboratories usually process the evidence samples at separate times and sometimes even different locations from the reference samples.

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TABLE 2.1 Typical DNA Amounts That May Be Extracted from Biological Materials (Lee & Ladd 2001). Both Quality and Quantity of DNA Recovered from Evidentiary Samples Can Be Significantly Affected by Environmental Factors.

Type of Sample	Amount of DNA
Liquid blood	20000 ng/mL to 40000 ng/mL
Blood stain	250 ng/cm^2 to 500 ng/cm^2
Liquid semen	150000 ng/mL to 300000 ng/mL
Post-coital vaginal swab	10 ng/swab to 3000 ng/swab
Plucked hair (with root)	1 ng/root to 750 ng/root
Shed hair (with root)	1 ng/root to 10 ng/root
Liquid saliva	1000 ng/mL to 10000 ng/mL
Oral swab	100 ng/swab to 1500 ng/swab
Urine	1 ng/mL to 20 ng/mL
Bone	3 ng/mg to 10 ng/mg
Tissue	50 ng/mg to 500 ng/mg

For many years, a popular method for preparation of reference samples was to make a blood stain by applying a drop of blood to a cotton cloth, referred to as a swatch, to produce a spot about 1 cm^2 in area. Ten microliters of whole blood, about the size of a drop, contains approximately 70,000 to 80,000 white blood cells and should yield approximately 500 ng of genomic DNA. The actual yield will vary with the number of white blood cells present in the sample and the efficiency of the DNA extraction process. The typical amounts of DNA extracted from various biological materials are shown in Table 2.1.

Extracted DNA is typically stored at −20°C, or even −80°C for long-term storage, to prevent nuclease activity. Nucleases are protein enzymes found in cells that degrade DNA to allow the cells to recycle nucleotide components. Nucleases need magnesium to work properly so one of the measures to prevent them from digesting DNA in blood is the use of purple-topped tubes containing a blood preservative known as EDTA. The EDTA chelates, or binds up, most of the free magnesium and thus helps prevent the nucleases from destroying the DNA in the collected blood sample.

PCR Inhibitors and DNA Degradation

When extracting biological materials for the purpose of forensic DNA typing, it is important to try to avoid further degradation of the DNA template as well as to remove inhibitors of PCR where possible. The presence of inhibitors or degraded DNA can lead to complete PCR amplification failure or a reduced sensitivity of detection usually for the larger PCR products.

Two PCR inhibitors commonly found in forensic cases are hemoglobin and indigo dyes from denim. Melanin found in hair samples can be a source of PCR inhibition when trying

to amplify mitochondrial DNA. These inhibitors likely bind in the active site of the *Taq* DNA polymerase and prevent its proper functioning during PCR amplification.

DNA degrades through a variety of mechanisms including both enzymatic and chemical processes. Once an organism dies its DNA molecules face cellular nucleases followed by bacterial, fungal, and insect onslaughts depending on the environmental conditions. In addition, hydrolytic cleavage and oxidative base damage can limit successful retrieval and amplification of DNA. The main target for hydrolytic cleavage is the glycosidic base sugar bond. Breakage here leads to nucleobase loss and then a single-stranded "nick" at the abasic site.

If a sufficient number of DNA molecules in the biological sample break in a region where primers anneal or between the forward and reverse primers, then PCR amplification (Chapter 4) efficiency will be reduced or the target region may fail to be amplified at all. Thus, heat and humidity, which speed up hydrolytic cleavage, are enemies of intact DNA molecules. Furthermore, UV irradiation (e.g., direct sunlight) can lead to cross-linking of adjacent thymine nucleotides on the DNA molecule which will prevent passage of the polymerase during PCR. The problems and potential solutions for samples possessing degraded or damaged DNA will be covered further in Chapter 10. In addition, methods for overcoming the effects of PCR inhibitors will be touched upon in Chapter 4.

EARLY TECHNIQUES USED FOR DNA EXTRACTION

There are several primary techniques for DNA extraction used in today's forensic DNA laboratory: organic extraction, Chelex extraction, and FTA or solid-phase extraction (Figure 2.1). The exact extraction or DNA isolation procedure varies depending on the type of biological evidence being examined. For example, whole blood must be treated differently from a bloodstain or a bone fragment. James Stray and colleagues at Applied Biosystems/Life Technologies have reviewed different DNA extraction methods (Stray et al. 2010, Stray & Shewale 2010).

Organic (Phenol-Chloroform) Extraction

Organic extraction, sometimes referred to as phenol chloroform extraction, has been in use for the longest period of time and for many years was the most widely used method for DNA extraction. High molecular weight DNA, which was essential for early RFLP methods, may be obtained most effectively with organic extraction. In the past decade, new extraction methods have been developed that use chemicals that are far less toxic than phenol.

Organic extraction involves the serial addition of several chemicals. First sodium dodecylsulfate (SDS) and proteinase K are added to break open the cell membranes and to break down the proteins that protect the DNA molecules while they are in chromosomes. Next a phenol/chloroform mixture is added to separate the proteins from the DNA. The DNA is more soluble in the aqueous portion of the organic–aqueous mixture. When centrifuged, the unwanted proteins and cellular debris are separated away from the aqueous phase and double-stranded DNA molecules can be cleanly transferred for analysis.

Some initial protocols involved a Centricon 100 (Millipore, Billerica, MA) dialysis and concentration step in place of the ethanol precipitation to remove heme inhibitors (Comey

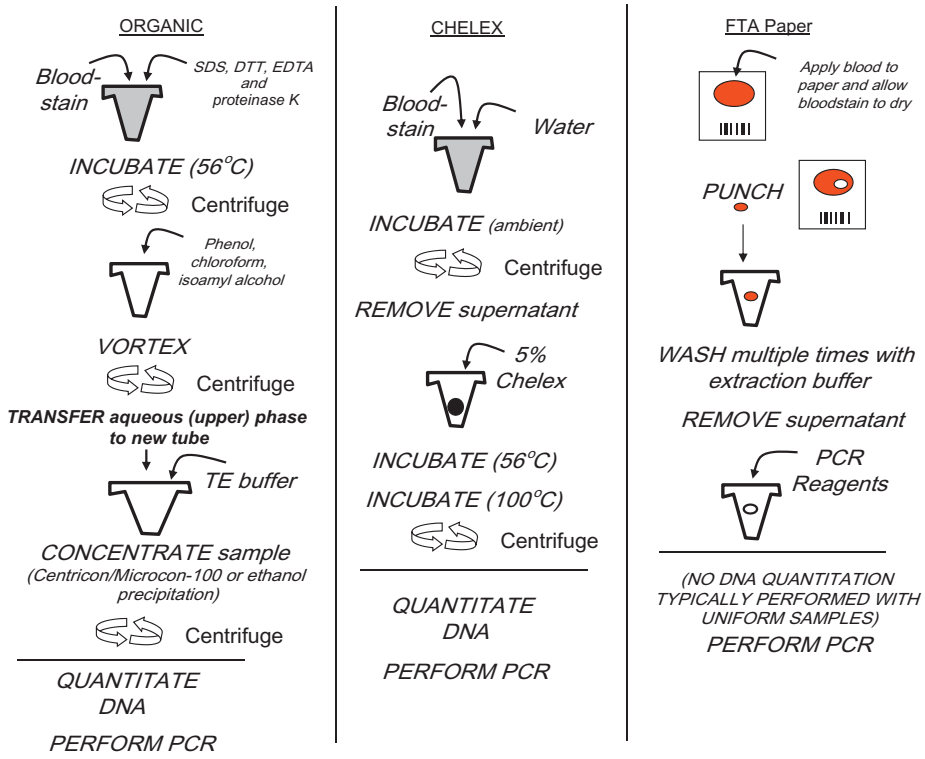


FIGURE 2.1 Schematic of commonly used DNA extraction processes.

et al. 1994). While the organic extraction method works well for recovery of high molecular weight DNA, it is time-consuming, involves the use of hazardous chemicals, and requires the sample to be transferred between multiple tubes, which increases the risk of error or contamination. When Millipore stopped selling Centricon 100 concentrators a few years ago, forensic DNA labs changed to Vivacon (Sartorius-Stedim, Concord, CA) or Microcon 100 (Millipore) ultrafiltration devices for sample concentration.

Chelex Extraction

An alternative and inexpensive procedure for DNA extraction that has become popular among forensic scientists is the use of a chelating-resin suspension that can be added directly to the sample (e.g., blood, bloodstain, or semen). Introduced in 1991 to the forensic DNA community, Chelex 100 (Bio-Rad Laboratories, Hercules, CA) is an ion-exchange resin that is added as a suspension to the samples (Walsh et al. 1991). The Chelex method of DNA extraction is more rapid than the organic extraction method. In addition, Chelex extraction involves fewer steps and thus fewer opportunities for sample contamination.

Chelex is composed of styrene divinylbenzene copolymers containing paired iminodiacetate ions that act as chelating groups in binding polyvalent metal ions such as magnesium.

Like iron filings to a magnet, the magnesium ions are drawn in and bound by the resin. By removing the magnesium from the reaction, DNA-destroying nuclease enzymes are inactivated and the DNA molecules are protected.

In most protocols, biological samples such as bloodstains are added to a 5% Chelex suspension and boiled for several minutes to break open the cells and release the DNA. An initial, prior wash step is helpful to remove possible contaminants and inhibitors such as heme and other proteins. The exposure to 100°C temperatures denatures the DNA as well as disrupting the cell membranes and destroying the cell proteins. After a quick spin in a centrifuge to pull the Chelex resin and cellular debris to the bottom of the tube, the supernatant is removed and can be added directly to the PCR amplification reaction.

Chelex denatures double-stranded DNA and yields single-stranded DNA from the extraction process. Thus, it can only be followed by PCR-based analyses. However, Chelex extraction is an advantage for PCR-based typing methods because it removes inhibitors of PCR and uses only a single tube for the DNA extraction, which reduces the potential for laboratory-induced contamination.

The addition of too much whole blood or too large a bloodstain to the Chelex extraction solution can result in some PCR inhibition. The AmpFlSTR kit manuals recommend 3 μL whole blood or a bloodstain approximately 3 mm × 3 mm (Applied Biosystems 1998).

FTA Paper

Another approach to DNA extraction involves the use of FTA paper. In the late 1980s, FTA paper was developed by Lee Burgoyne at Flinders University in Australia as a method for storage of DNA (Burgoyne et al. 1994). FTA originally stood for "Fitzco/Flinder Technology Agreement." FTA paper is an absorbent cellulose-based paper that contains four chemical substances to protect DNA molecules from nuclease degradation and preserve the paper from bacterial growth (Burgoyne 1996). As a result, DNA on FTA paper is stable at room temperature over a period of several years. However, a 2002 study evaluating FTA and three other commercial papers as DNA storage media found little difference in their ability to obtain typeable STR results after 19 months of storage (Kline et al. 2002).

Use of FTA paper simply involves adding a spot of blood to the paper and allowing the stain to dry. The cells are lysed upon contact with the paper and DNA from the white blood cells is immobilized within the matrix of the paper. A small punch of the paper is removed from the FTA card bloodstain and placed into a tube for washing. The bound DNA can then be purified by washing it with FTA Purification Reagent (Whatman, Clifton, NJ) to remove heme and other inhibitors of the PCR reaction. This purification of the paper punch can be seen visually because as the paper is washed, the hemoglobin red color is removed with the supernatant. The clean punch is then added directly to the PCR reaction. Alternatively, some groups have performed a Chelex extraction on the FTA paper punch and used the supernatant in the PCR reaction (Lorente et al. 1998, Kline et al. 2002).

Devices have also been developed for collection of saliva or buccal cells using a spongy swab that is then pressed against an FTA card to transfer the collected cells for sample preservation. Indicator paper that changes color with liquid contact is typically used to enable visualization of sample transfer to the FTA card.

A major advantage of FTA paper is that consistent results may be obtained without quantification because a uniform amount of cells are typically being sampled. Furthermore the procedure may be automated on a robotic workstation (Belgrader et al. 1995, Belgrader & Marino 1997, Tack et al. 1997). For situations where multiple assays need to be run on the same sample, a bloodstained punch may be reused for sequential DNA amplifications and typing (Del Rio et al. 1996). Unfortunately, due to static electricity, dry paper punches do not like to stay in their assigned tubes and can "jump" between wells in a sample tray. Thus, this method is not as widely used today as was once envisioned. However, due to its preservation and storage capabilities, efforts have been made to use FTA cards for more widespread collection of crime scene evidence (Lorente et al. 2004).

Other Methods

A few other methods have been used for DNA extraction from biological samples including "salting out" the DNA with sodium acetate (Miller et al. 1988), use of sodium hydroxide (Klintschar & Neuhuber 2000), and digestion of cellular proteins with a thermostable proteinase (Moss et al. 2003). A liquid extraction method using the thermostable proteinase approach has been developed by ZyGEM (Hamilton, New Zealand).

SOLID-PHASE DNA EXTRACTION METHODS

With the desire to automate more steps in DNA analysis, many laboratories have moved to various forms of solid-phase extraction where DNA is selectively bound to a substrate such as silica particles. The DNA is retained while proteins and other cellular components are washed away. Then the DNA is released in a purified form. Chapter 17 will discuss more on specific robotic platforms for automated sample processing. The bind-wash-elute format is illustrated in Figure 2.2.

The most widely used solid-phase extraction methods are QIAGEN columns, DNA IQ, and PrepFiler. The primary characteristics of each method are featured in Table 2.2.

QIAGEN Extraction Chemistry and Kits

Solid-phase extraction methods for DNA have been developed in recent years in formats that enable high-throughput DNA extractions. One of the most active efforts in this area is with silica-based extraction methods and products from QIAGEN, Inc. (Valencia, CA). For more than a decade, QIAamp spin columns have proven effective as a means of DNA isolation (Greenspoon et al. 1998).

In this approach, nucleic acids selectively absorb to silica on a support, such as small glass beads, in the presence of high concentrations of chaotropic salts such as guanidine hydrochloride, guanidine isothiocyanate, sodium iodide, and sodium perchlorate (Vogelstein & Gillespie 1979, Boom et al. 1990, Duncan et al. 2003). These chaotropic salts disrupt hydrogen-bonding networks in liquid water and thereby make denatured proteins and nucleic acids more thermodynamically stable than their native folded or structured counterparts (Tereba et al. 2004).

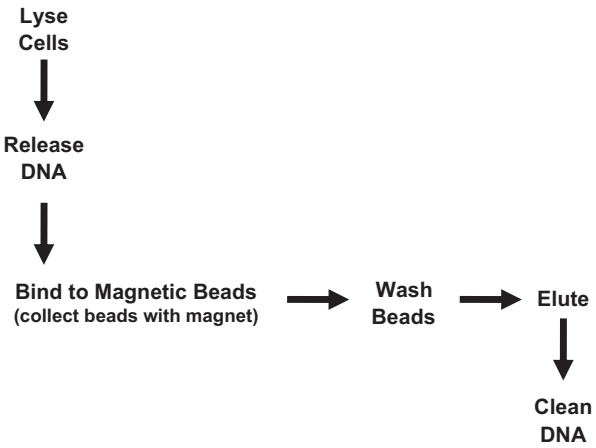


FIGURE 2.2 Bind-wash-elute method utilized in DNA IQ and PrepFiler DNA extraction and purification chemistries.

TABLE 2.2 Comparison of DNA Extraction Methods.

	QIAGEN	Promega	Applied Biosystems
Products	QIAamp	DNA IQ	PrepFiler
Solid-phase	Silica beads	Magnetic silica beads	Magnetic silica beads
Wash performed	Centrifugation or vacuum manifold	Magnet	Magnet
Robotic platform	EZ1, M48, QIACube	Maxwell 16, Tecan	AutoMate Express, Tecan

If the solution is more acidic than pH 7.5, DNA adsorption to the silica is typically around 95% and unwanted impurities can be washed away. Under alkaline conditions and low salt concentrations, the DNA will efficiently elute from the silica material. This solid-phase extraction approach can be performed with centrifugation or vacuum manifolds in single tube or 96-well plate formats (Hanselle et al. 2003, Yasuda et al. 2003) and is even being developed into formats that will work on a microchip (Wolfe et al. 2002, Bienvenue & Landers 2010). Several robotic platforms have been developed to enable automated processing of QIAGEN DNA extractions including the EZ1, M48, and QIAcube. The EZ1 liquid-handling robot (Figure 2.3) can process either 6 or 14 samples (depending on the model) in about 20 minutes following a previous incubation/cell lysis step.

DNA IQ

Another solid-phase extraction approach is the DNA IQ system marketed by Promega Corporation (Madison, WI). The DNA IQ system, which stands for "isolation" and



FIGURE 2.3 QIAGEN EZ1 DNA extraction robot.

"quantitation," utilizes the same silica-based DNA binding and elution chemistries as QIAGEN kits but with silica-coated paramagnetic resin (Tereba et al. 2004). With this approach, DNA isolation can be performed in a single tube by simply adding and removing solutions.

First, the DNA molecules are reversibly bound to the magnetic beads in solution with a pH more acidic than pH 7.5 (the same as noted previously for the QIAGEN chemistry). A magnet is used to draw the silica-coated magnetic beads to the bottom or side of the tube leaving any impurities in solution. These solution impurities (proteins, cell debris, etc.) are easily removed by drawing the liquid off of the beads. The magnetic particles with DNA attached can be washed multiple times to more thoroughly clean the DNA. Finally, a defined amount of DNA can be released into solution via heating for a few minutes.

The quantity of DNA isolated with this approach is based on the number and capacity of the magnetic particles used. Since flow-through vacuum filtration or centrifugation steps are not used, magnetic bead procedures enable simple, rapid, and automated methods. This extraction method has been automated on the Beckman 2000 robot workstation and implemented into forensic casework by the Virginia Department of Forensic Science and a growing number of other crime labs (Greenspoon et al. 2004).

A Maxwell 16 robot (Promega Corporation) is often used to automate the DNA IQ extraction sample processing. The Maxwell 16 can extract 16 samples simultaneously at a price of about $6 per sample in less than 30 minutes following off-instrument incubation. For laboratories needing higher throughput, Tecan liquid-handling robots have been used (Frégeau et al. 2010). Methods are also being developed to simultaneously extract both DNA and RNA using the DNA IQ chemistry (Bowden et al. 2010).

The magnetic bead approach is dependent on DNA binding to the beads. If anything interferes with this binding, the DNA sample can be lost in the washes. During part of a large criminal investigation involving processing of tens of thousands of samples, the Royal

Canadian Mounted Police (RCMP) DNA lab found that chemicals present in a presumptive test for blood known as Hemastix prevented the expected DNA extraction results with DNA IQ (Poon et al. 2009). Additional studies by the RCMP found that chemical warfare agents such as sulfur mustard, sarin, sodium 2-fluoroacetate, and diazinon were removed to below detectable limits as part of the DNA IQ extraction process (Wilkinson et al. 2007a, 2007b). Thus, the DNA IQ process can remove these dangerous chemicals and render samples safe to handle following DNA extraction.

PrepFiler

In 2008 Applied Biosystems released a magnetic particle based DNA extraction technology named PrepFiler that is similar to DNA IQ (Applied Biosystems 2008). PrepFiler has been validated (Barbaro et al. 2009, Brevnov et al. 2009) and enables isolation of high-quality DNA from forensic samples in high yields primarily because a small bead size is used to provide a higher surface area (compared to larger magnetic bead systems) for capturing DNA molecules during the extraction process.

Up to 96 samples can be processed at a time in under 2.5 hours using the Tecan Freedom EVO automated liquid-handling workstation. A small liquid-handling robot similar to the QIAGEN EZ1 has also been released by Applied Biosystems called the AutoMate Express Forensic DNA Extraction System (Applied Biosystems 2010). To improve extraction efficiencies with calcified tissues including bone and teeth as well as adhesives, Applied Biosystems has developed a BTA (bone, teeth, adhesive) lysis buffer protocol (Stray et al. 2009).

Protocols for Various Tissue Types

Optimal DNA extraction may come from slightly different procedures depending on the source of the biological sample. The reference lists at the back of this chapter include citations to methods developed for DNA extraction from a variety of tissue types including bloodstains, bone, formalin-fixed paraffin embedded (FFPE) tissues, hair, teeth, urine, and saliva from stamps and envelopes.

DIFFERENTIAL EXTRACTION

Differential extraction is a modified version of the organic extraction method that separates epithelial and sperm cells (Figure 2.4). Differential extraction was first described in 1985 by Peter Gill and coworkers (Gill et al. 1985). This method is commonly used today by the FBI Laboratory and other forensic crime laboratories to isolate the female and male fractions in sexual assault cases that contain a mixture of male and female DNA. By separating the male fraction away from the victim's DNA profile, it is much easier to interpret the perpetrator's DNA profile in a rape case.

The differential extraction procedure involves preferentially breaking open the female epithelial cells with incubation in an SDS/proteinase K mixture. Sperm nuclei are subsequently lysed by treatment with an SDS/proteinase K/dithiothreitol (DTT) mixture. The DTT breaks down the protein disulfide bridges that make up sperm nuclear membranes. Differential

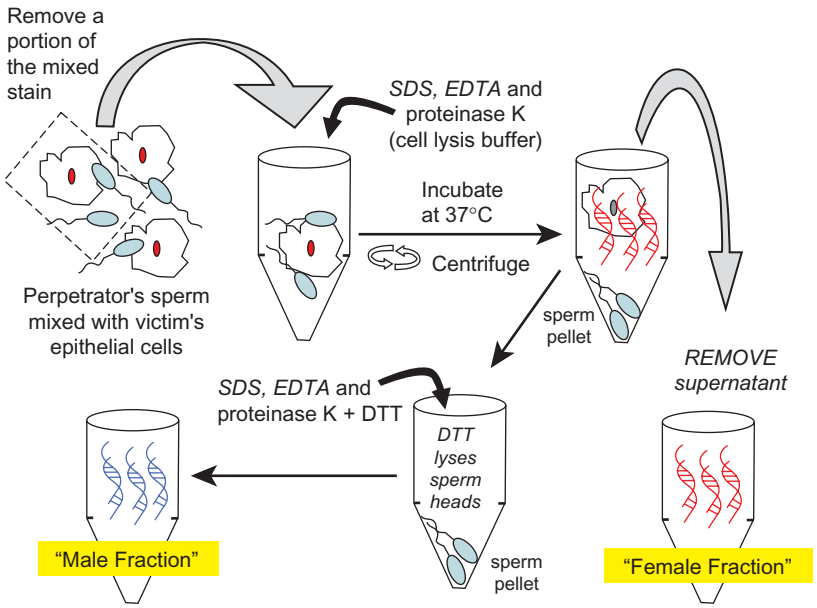


FIGURE 2.4 Schematic of differential extraction process used to separate male sperm cells from female epithelial cells.

extraction works because sperm nuclei are impervious to digestion without DTT. The major difference between the regular version of organic extraction described earlier and differential extraction is the initial incubation in SDS/proteinase K without DTT present.

Promega Corporation (Madison, WI) has developed an automated Differex method that involves using DNA IQ magnetic beads to hold the sperm pellet in place while a separation solution keeps the digestion buffer and epithelial DNA away from the sperm pellet during the wash steps (Knox & Olsen 2008). Up to 40 samples can be processed in a 96-well plate in less than 5 hours using Differex.

Differential extraction works well in most sexual assault cases to separate female and male fractions from one another (Figure 2.5). However, some perpetrators of sexual assaults have had a vasectomy in which case there is an absence of spermatozoa. Azoospermic semen, i.e., without sperm cells, cannot be separated from the female fraction with differential extraction. In the case of azoospermic perpetrators, the use of Y chromosome specific markers permit male DNA profiles to be deduced in the presence of excess female DNA. Failure to separate the male and female portions of a sexual assault sample results in a mixture of both the perpetrator's and the victim's DNA profiles. DNA mixtures can sometimes be challenging to interpret.

Other Approaches

Figure 2.6 compares several other approaches for separating victim and perpetrator cells and/or DNA in sexual assault evidence. Methods for directly capturing sperm cells have

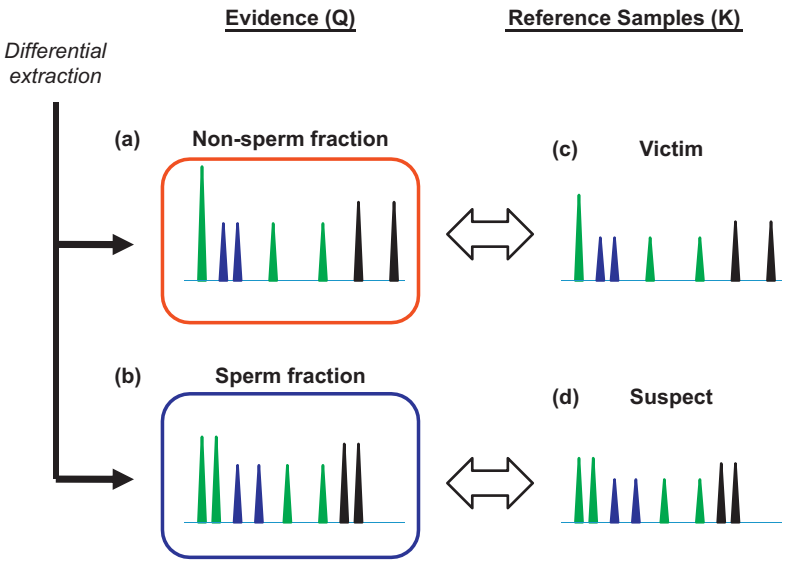


FIGURE 2.5 Illustration of results from the four samples typically associated with a sexual assault case. During differential extraction, the evidence sample is divided into two fractions: (a) non-sperm or epithelial fraction and (b) the sperm fraction (see Figure 2.4). The suspect result (d) is compared to the sperm fraction (b) to see if they match. The victim's DNA profile (c) provides confirmation of sample integrity with the non-sperm fraction (a).

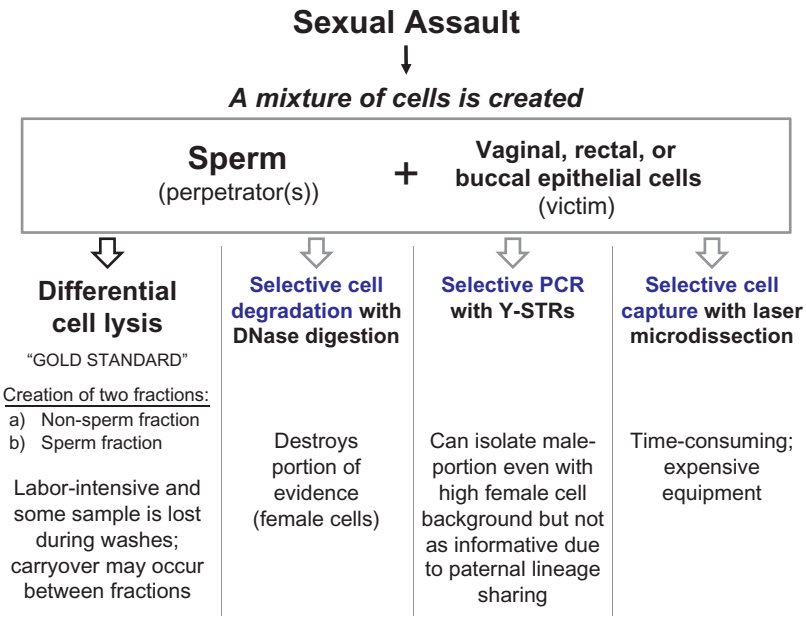


FIGURE 2.6 Different methods for separating perpetrator(s)'s sperm and victim's epithelial cells created in sexual assault DNA samples.

been developed including laser-capture microdissection (see below). By physically separating the perpetrator's sperm cells from the victim's epithelial cells, the perpetrator's DNA can be enriched and isolated from even a vast preponderance of victim's cells. Digestion of victim's epithelial cells to try to increase yield from the perpetrator's DNA has also been successful (Gavin et al. 2009).

Laser-Capture Microdissection

Sperm cells can be selectively captured using a clinical procedure known as *laser-capture microdissection* (LCM), which is commonly used to select tumor cells from surrounding tissue on microscope slides. Sperm cells from sexual assault evidence spread on microscope slides can be collected with laser-capture microdissection to perform reliable STR testing (Elliott et al. 2003, Sanders et al. 2006). When sperm cells are observed in the field of view of the microscope, a tiny laser is activated and a thin plastic film placed over the slide melts at the specific point of laser light contact to capture or enclose the cell of interest. By moving the microscope slide around, dozens of sperm cells are collected onto this thin film that sits directly above the sample. The collection film is then transferred to a tube where DNA from the isolated sperm can be extracted and amplified using the polymerase chain reaction. Other LCM methods catapult identified cells directly into a collection tube.

Non-sperm male cells may also be detected with *fluorescent in-situ hybridization* (FISH) techniques enabling selective capture of male versus female cells from a mixed cell population (Anslinger et al. 2007). While the use of FISH and LCM add to the expense of forensic casework, these clinical techniques can enable speciality cases to be solved that would not otherwise be possible.

DIRECT PCR TO BYPASS DNA EXTRACTION

One of the latest developments in forensic DNA typing is the application of direct PCR with no DNA extraction. Improved reaction components and engineered DNA polymerases permit PCR amplification without DNA extraction and purification steps to remove inhibitors (see Chapter 4). The PowerPlex 16 HS and Identifiler Direct STR typing kits (see Chapter 5) enable bloodstains from reference samples to be processed more rapidly without DNA extraction. An advantage of this approach is that samples are effectively concentrated as DNA material is not lost during wash steps that typically occur with DNA extraction methods.

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